Question 8		
(a)	Ther rate of decay is not affected by external factors like temperature and pressure nor by external stimuli.	**B1**
(b)(i)	Alpha radiation is unsuitable because its range in human tissue is too short, meaning it cannot penetrate deeply enough to destroy the entire tumour mass.	**B1**
(b)(ii)1.	Number of iodine particles = $\frac{1.2\times10^{-9}}{131}\times6.02\times10^{23}=5.5\times10^{12}$ Activity = λN Activity $=\frac{8.6\times10^{-2}}{24\times3600}\times(5.5\times10^{12})$ $=5.5\times10^{6}$ Bq	**C1** **C1** **A1**
(b)(ii)2.	$A=A_0e^{-\lambda t}$ $\frac{1}{50}=e^{-\frac{8.6\times10^{-2}}{24\times3600}t}$ $t=3.9\times10^{6}$ s	**C1** **A1**